Stokes glacier models, with Firedrake

a finite element tutorial

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version 2.2 (September 2026)

github.com/bueler/stokes-ice-tutorial

making this presentation effective

- please ask questions, thanks!
- send questions or comments:
  - by email to elbueler@alaska.edu
  - or by using issues at github.com/bueler/stokes-ice-tutorial/

- these slides are the primary documentation for some Python codes
- please try the codes, and give me your feedback?

Outline

theory	equations for the glacier dynamics problem
`stage1/`	first demo: linear Stokes
`stage2/`	power-law Stokes for glacier ice
`stage3/`	more power: extruded meshes and 3D glaciers
theory	evolving glaciers: free-surface equation coupled to Stokes
`stage4/`	moving-margin dome solutions
	summary and learning more

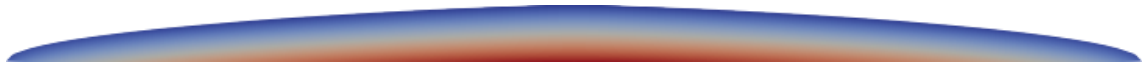
`github.com/bueler/stokes-ice-tutorial`

glacier dynamics using a Stokes model: the high-level view

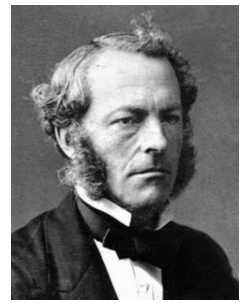
- we apply conservation principles to glaciers:
  - mass conservation ✓
  - momentum conservation ✓
  - energy conservation (not here)
- incompressibility is one aspect of mass conservation
  - another is the free-surface equation . . . see `stage4/`
- glaciers are a "slow flow" problem because inertial forces are tiny
  - momentum conservation is therefore called a "stress balance"
  - velocity & pressure are instantaneously determined by glacier geometry and boundary stresses

velocity

pressure



- Stokes equations = stress balance + incompressibility



George Stokes

Glen-Steinemann-Nye-Stokes equations

- the canonical glacier dynamics equations are from Glen, Steinemann, Nye, and Stokes; the model is isotropic *power-law Stokes*
- the main equations:

$$-\nabla \cdot \tau + \nabla p = \rho \mathbf{g} \qquad \textit{stress balance}$$

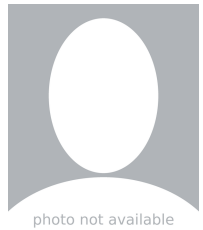
$$\nabla \cdot \mathbf{u} = 0 \qquad \textit{incompressibility}$$

$$\tau = B_n |D\mathbf{u}|^{(1/n)-1} D\mathbf{u} \qquad \textit{flow law}$$

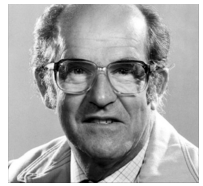
  - $\mathbf{u}$ is velocity, p is pressure, τ is the deviatoric stress tensor, and $D\mathbf{u} = \frac{1}{2}\left(\nabla\mathbf{u} + \nabla\mathbf{u}^\top\right)$ is the strain-rate tensor
  - our model has *constant ice density*; incompressibility follows

- constants (*with default values in demos*):
  - $\rho = 910\,\mathrm{kg\,m^{-3}}$
  - $|\mathbf{g}| = 9.81\,\mathrm{m\,s^{-2}}$
  - $n = 3$
  - $B_n = 6.8 \times 10^7\,\mathrm{Pa\,s^{1/3}}$ ← *constant because isothermal here*

John Glen



Samuel G. Steinemann (1923–2016)



John Nye

viscosity, and avoiding unbounded values

- the classical viscosity relationship $\tau = 2\nu D\mathbf{u}$ eliminates τ from the equations, giving a system for $\mathbf{u}, p$
- however, the unmodified power-law viscosity formula yields unbounded values as $D\mathbf{u} \to 0$, i.e. for vanishing strain rates
  - note that "$(1/n) - 1$" in $\nu = \frac{1}{2} B_n |D\mathbf{u}|^{(1/n)-1}$ is a negative power
- so we also regularize with small $\epsilon > 0$ to generate bounded viscosity

power-law Stokes equations with regularized viscosity

$$-\nabla \cdot (2\nu_\epsilon(|D\mathbf{u}|) D\mathbf{u}) + \nabla p = \rho \mathbf{g} \qquad \textit{stress balance}$$

$$\nabla \cdot \mathbf{u} = 0 \qquad \textit{incompressibility}$$

$$\nu_\epsilon(|D\mathbf{u}|) = \tfrac{1}{2} B_n \left(|D\mathbf{u}|^2 + (\epsilon\, D_0)^2 \right)^{((1/n)-1)/2} \qquad \textit{flow law}$$

  - we use $\epsilon = 10^{-4}$ and $D_0 = 1\,\mathrm{a}^{-1}$ in the demonstrations

boundary conditions, as simple as possible

- note that $\sigma = 2\nu_\epsilon D\mathbf{u} - pI$ is the Cauchy stress tensor
- stress-free top:

$$\sigma\mathbf{n} = (2\nu_\epsilon D\mathbf{u} - pI)\,\mathbf{n} = \mathbf{0}$$

  - atmospheric pressure could be applied at surface . . . no interesting effect
- no-slip base:

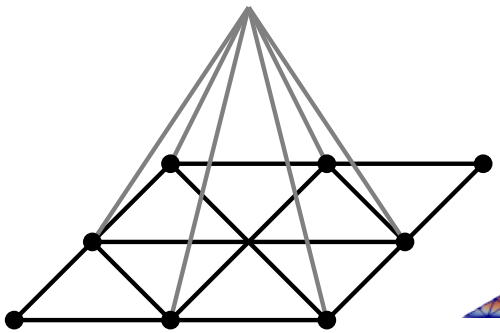
$$\mathbf{u} = \mathbf{0}$$

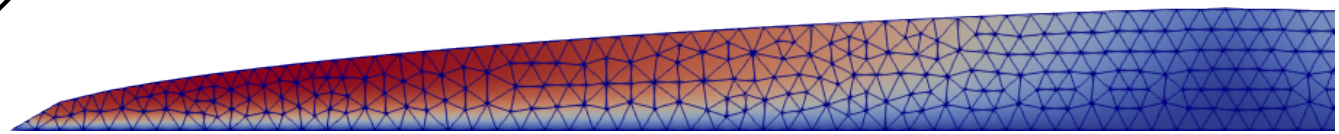
  - no attempt to model basal sliding in this tutorial

speed $|\mathbf{u}|$

finite elements . . . with no details

- by-hand solution of the Stokes model is impossible for most geometries
  - *exercise.* solve slab-on-a-slope, which is an exception
- so we solve numerically using the finite element (FE) method
- the FE method allow us to start from a mesh of triangles (or quadrilaterals, prisms, . . .) over our domain, then express the approximate solution in terms of functions built from the mesh, but I will skip the details . . .





- the whole point of Firedrake is suppress the details of FE!
  - but try Elman, Sylvester & Wathen (2014) for self-study?
- however, one needs to know a key step: write a weak form

the weak form

- start from the equations, now called the *strong form*:

$$-\nabla \cdot (2\nu_\epsilon(|D\mathbf{u}|)\, D\mathbf{u}) + \nabla p = \rho \mathbf{g} \tag{1}$$

$$\nabla \cdot \mathbf{u} = 0 \tag{2}$$

- multiply (1) by test velocity $\mathbf{v}$ and (2) by test pressure q, integrate by parts, and combine into one nonlinear residual function:

$$F(\mathbf{u}, p; \mathbf{v}, q) = \int_\Lambda 2\nu_\epsilon(|D\mathbf{u}|)\, D\mathbf{u} \,:\, D\mathbf{v} - p\nabla \cdot \mathbf{v} - q\nabla \cdot \mathbf{u} - \rho \mathbf{g} \cdot \mathbf{v}\, dx$$

  - Λ is the spatial domain of ice
  - "dx" $=$ $dx\, dy\, dz$ in 3D, and "dx" $=$ $dx\, dz$ in the planar case

- the actual weak-form problem statement is "residual $= 0$":

$$F(\mathbf{u}, p; \mathbf{v}, q) = 0 \qquad \text{for all test functions } \mathbf{v} \text{ and } q$$

  - the FE method solves the same problem, but over a finite-dimensional space

UFL for the weak form

- nonlinear residual function from last slide:

$$F(\mathbf{u}, p; \mathbf{v}, q) = \int_\Lambda 2\nu_\epsilon(|D\mathbf{u}|)\, D\mathbf{u} \,:\, D\mathbf{v} - p\nabla\cdot\mathbf{v} - q\nabla\cdot\mathbf{u} - \rho\mathbf{g}\cdot\mathbf{v}\, dx$$

  with $\nu_\epsilon(|D\mathbf{u}|) = \frac{1}{2}B_n\left(|D\mathbf{u}|^2 + (\epsilon\, D_0)^2\right)^{((1/n)-1)/2}$

- we write this using the *Unified Form Language* (UFL) in Firedrake:

```
def D(w):
    return 0.5 * (grad(w) + grad(w).T)

Du2 = 0.5 * inner(D(u), D(u)) + (eps * Dtyp)**2.0
nu = 0.5 * Bn * Du2**((1.0 / n - 1.0)/2.0)

fbody = rho * Constant((0.0, 0.0, - g))
F = ( 2.0 * nu * inner(D(u), D(v)) \
      - p * div(v) - q * div(u) - inner(fbody, v) ) * dx
```

  - Python UFL code is quite similar to the math

mixed finite elements for fluid problems

- other details we need to know about:
  1. choose separate function spaces for the velocity and the pressure
  2. choose finite element spaces that (the experts say) are stable

- Firedrake makes choosing such mixed elements easy:

```
V = VectorFunctionSpace(mesh, 'CG', 2)  # u space is P2
W = FunctionSpace(mesh, 'CG', 1)        # p space is P1
Z = V * W                               # mixed space
up = Function(Z)
u, p = split(up)
v, q = TestFunctions(Z)
```

  - this $P_2 \times P_1$ choice is called Taylor-Hood
  - other mixed spaces are just as easy
  - we choose $P_2 \times DQ_1$ in `stage4/`, for reasons . . .

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theory equations for the glacier dynamics problem

`stage1/` first demo: linear Stokes

`stage2/` power-law Stokes for glacier ice

`stage3/` more power: extruded meshes and 3D glaciers

theory evolving glaciers: free-surface equation coupled to Stokes

`stage4/` moving-margin dome solutions

summary and learning more

`github.com/bueler/stokes-ice-tutorial`

the linear Stokes equations

- if we make viscosity constant ($\nu_0 > 0$) then we get a linear Stokes system:

$$-\nabla \cdot (2\nu_0\, D\mathbf{u}) + \nabla p = \rho \mathbf{g}$$
$$\nabla \cdot \mathbf{u} = 0$$

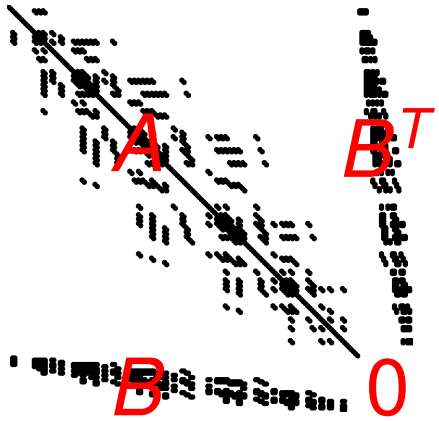
- classical reformat with $2\nabla \cdot D\mathbf{u} = \nabla^2 \mathbf{u}$:

$$-\nu_0 \nabla^2 \mathbf{u} + \nabla p = \rho \mathbf{g}$$
$$-\nabla \cdot \mathbf{u} = 0$$

- which has symmetric block structure:

$$\begin{bmatrix} A & B^\top \\ B & 0 \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ p \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix}$$

  - $A = -\nu_0 \nabla^2 \geq 0$
  - $B^\top = (-\nabla \cdot)^\top = \nabla$ (integration-by-parts)



running the demonstration programs

- let's run a demo
- install Firedrake following these directions:

  firedrakeproject.org/install.html
- activate the Python virtual environment every time you use Firedrake:

```
$ source ~/venv-firedrake/bin/activate
```

- other tools you will need:
  - Gmsh gmsh.info
  - Paraview paraview.org
- clone this tutorial, both for the codes and these slides:

```
$ git clone https://github.com/bueler/stokes-ice-tutorial.git
```

linear Stokes

- *purpose:* solve linear Stokes on a trapezoidal “glacier”
- *source files:* `trap.geo`, `solve.py` ← inspect these
- *generated files:* `trap.msh`, `trap.pvd`

```
$ cd stage1/
$ gmsh -2 trap.geo          # mesh the domain
$ gmsh trap.msh             # view the mesh
$ python3 solve.py          # solve linear Stokes
$ paraview trap.pvd         # view the solution
```

speed $|\mathbf{u}|$

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Newton's method

- linear Stokes in `stage1/` solves a single linear system:

$$\begin{bmatrix} A & B^\top \\ B & 0 \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ p \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix}$$

  - default solver: Firedrake $\to$ PETSc KSP $\to$ MUMPS
- but Glen-Stokes residual function is nonlinear in $\mathbf{u}$:

$$F(\mathbf{u}, p; \mathbf{v}, q) = \int_\Lambda 2\,\nu_\epsilon(|D\mathbf{u}|)\,D\mathbf{u} \,:\, D\mathbf{v} - p\nabla\cdot\mathbf{v} - q\nabla\cdot\mathbf{u} - \rho\mathbf{g}\cdot\mathbf{v}\,dx$$

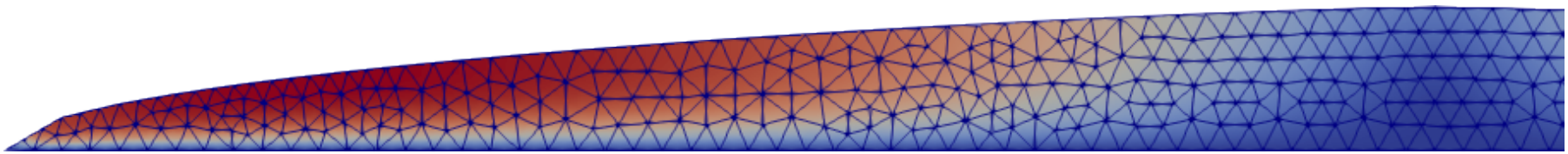
- fortunately, Firedrake applies Newton's method by default if we write the problem as `solve(F == 0,...)`
- what you need to know about Newton's iteration:
  - it does linearization around each iterate
  - Firedrake uses UFL symbolic differentiation for linearization
  - Firedrake asks PETSc's SNES to do the Newton iteration
  - SNES options will monitor and control the iteration

Isaac Newton

power-law Stokes for a glacier

- *purpose:* solve the power-law Stokes equations on a dome-shaped glacier with a flat bed
- *source files:* `domain.py`, `solve.py` ← inspect these
- *generated files:* `dome.geo`, `dome.msh`, `dome.pvd`

```
$ cd stage2/
$ python3 domain.py      # generate domain outline
$ gmsh -2 dome.geo       # mesh the domain
$ python3 solve.py       # solve power-law Stokes
$ paraview dome.pvd      # view the solution
```



speed $|\mathbf{u}|$

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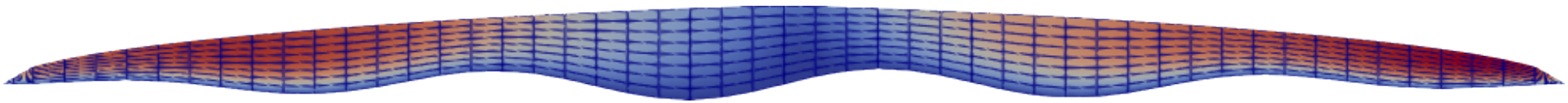
a more practical and powerful glacier solver

- this stage takes a more practical and powerful approach
- we switch to an extruded mesh, and allow a bumpy bed (below)
  - the base mesh has lower dimension
  - now the elements are quadrilaterals in the planar case, versus triangles in `stage2/`
- we demonstrate (mostly) dimension-independent programming
  - code for 2D and 3D glacier geometry is nearly the same; elements are prisms in 3D
  - dimension chosen at runtime
- additional bells & whistles:
  - rescaling the equations
  - vertical grid sequencing, with over-regularization on coarse meshes
  - additional diagnostic fields in the output `.pvd` file

solution speed $|\mathbf{u}|$ for default run:

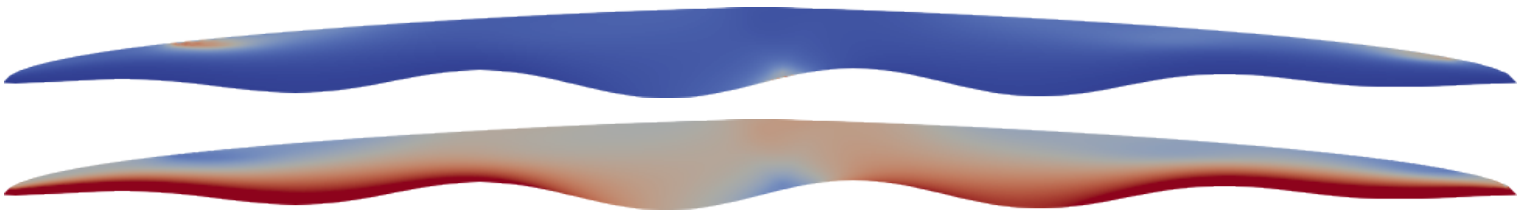


stage3/ extruded mesh and more

- *purpose:* a practical and robust solver
  - construct mesh by vertically extruding a base mesh — *practical geometry*
  - rescale the equations — *better conditioning*
  - vertical grid sequencing — *better initial iterates*
  - write ν_ϵ, τ, $p - p_{\text{hydrostatic}}$ into output file — *better diagnostics*
- *source file:* `solve.py` ← inspect this
- *generated file:* `dome.pvd`

```
$ cd stage3/
$ python3 solve.py                          # default resolution
$ python3 solve.py -mx 400 -refine 2        # higher resolution
$ paraview dome.pvd
```

effective viscosity ν_ϵ and deviatoric stress magnitude $|\tau|$:



evidence of convergence

- we can now set resolution at runtime, for example:

```
$ python3 solve.py -b0 0 -mx 400 -mz 4 -refine 2
```

  - using optional flat bed (`-b0 0`), to compare results to `stage2/`
  - initial mesh has 400 elements in horizontal and 4 in vertical
  - refine the vertical twice (by factors of four), to generate a 400×64 mesh

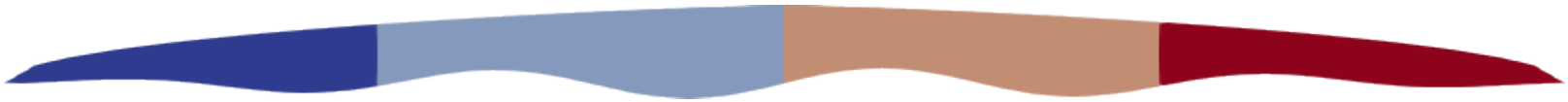
- velocity results, in meters per year, from a refinement study:

mesh	$\Delta x \times \Delta z$ (m)	average $\lvert\mathbf{u}\rvert$	maximum $\lvert\mathbf{u}\rvert$
50×8	400×125	1782	3263
100×16	200×63	1766	3211
200×32	100×31	1761	3196
400×64	50×16	1759	3191
800×128	25×8	1759	3190

  - velocities are apparently converging, but we do not know the exact solution . . . this is not verification

parallel computation

- as usual with Firedrake, everything works in parallel
  - extruded mesh gets partitioned following base mesh partition
  - see `rank` field in `.pvd` output; result with 4 processes:



- using the default direct solver (Mumps), results are essentially independent of number of processes
- however, speed-up is sublinear because of significant serial workloads

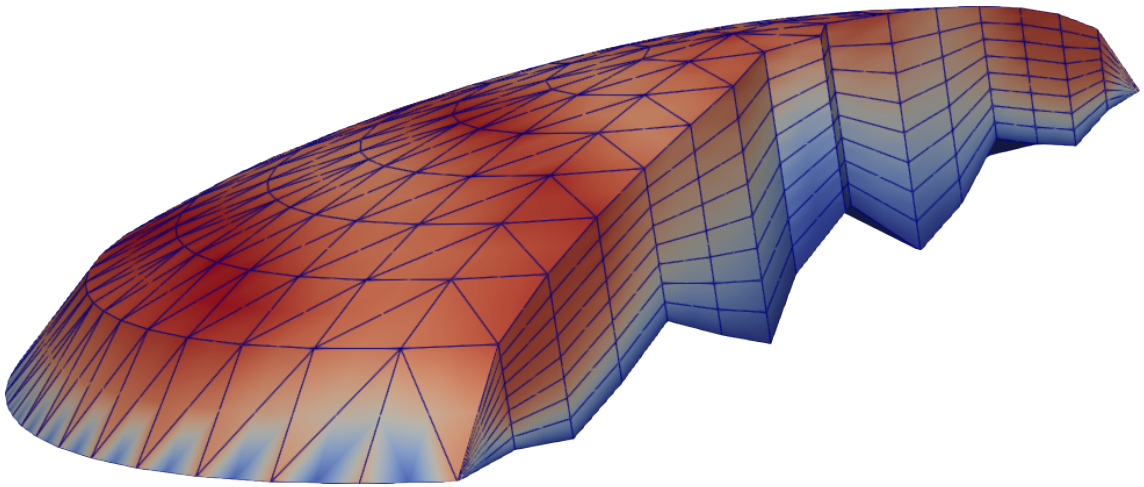
```
$ python3 solve.py -mx 800 -refine 2                # 38 secs run time

$ mpiexec -n 4 python3 solve.py -mx 800 -refine 2   # 20 secs run time
```

- your performance results may vary . . .

`stage3/` continued a 3D glacier

- *purpose:* demonstrate a (mostly) dimension-independent Firedrake code, to model a 3D glacier
- *source file:* `solve.py` ← inspect this
- *generated file:* `dome.pvd`

```
$ python3 solve.py -dim 3 -baserefine 3 -refine 1
$ paraview dome.pvd
```



3× vertical exaggeration, coloring by speed $|\mathbf{u}|$

solver performance limitations

- this run is about as far as I can refine on my office desktop:

```
$ mpiexec -n 12 python3 solve.py -dim 3 -baserefine 4 \
    -refine 2 -s_snes_atol 1.0e-2 -o finest.pvd
```

- *limiting factor:* the direct solver for each Newton step linear system uses too much memory when generating the LU factors
  - direct solvers experience *fill-in*, especially in 3D; see B (2021b)
- for higher resolution we need a scalable solver
  - such a solver would be built by composition of features: matrix-free, Schur-complement preconditioning, multigrid on the **u**-**u** block; see B (2021b)
  - for a scalable implementation see Isaac, Stadler, & Ghattas (2015)
  - addressing scalability properly is a different talk, but see B (2023)

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but don't glaciers actually change with the climate?

- we do not yet have an evolving glacier model
  - time derivatives have not even appeared!
  - the climate (e.g. surface mass balance) has not appeared as an input

Q. how do glaciers change shape?

A. via the free-surface equation, a kinematical and mass conservation statement:

$$\frac{\partial s}{\partial t} - \mathbf{u}|_s \cdot \mathbf{n}_s = a$$

  - $z = s(t, x, y)$ is the surface elevation of the glacier
  - $\mathbf{u}|_s$ is the surface value (*trace*) of the ice flow velocity
  - $\mathbf{n}_s = \left\langle -\frac{\partial s}{\partial x}, -\frac{\partial s}{\partial y}, 1 \right\rangle$ is a normal vector to the surface
  - $a(t, x, y)$ is the surface mass balance, the source term

- equation holds on $(0, T) \times \Omega$, where $\Omega \subset \mathbb{R}^d$ is a fixed map-plane region
- initial surface $s(0, x, y)$ is needed

coupled equations

coupled free-surface & power-law Stokes equations

$$\frac{\partial s}{\partial t} - \mathbf{u}|_s \cdot \mathbf{n}_s = a \qquad \textit{free-surface equation}$$

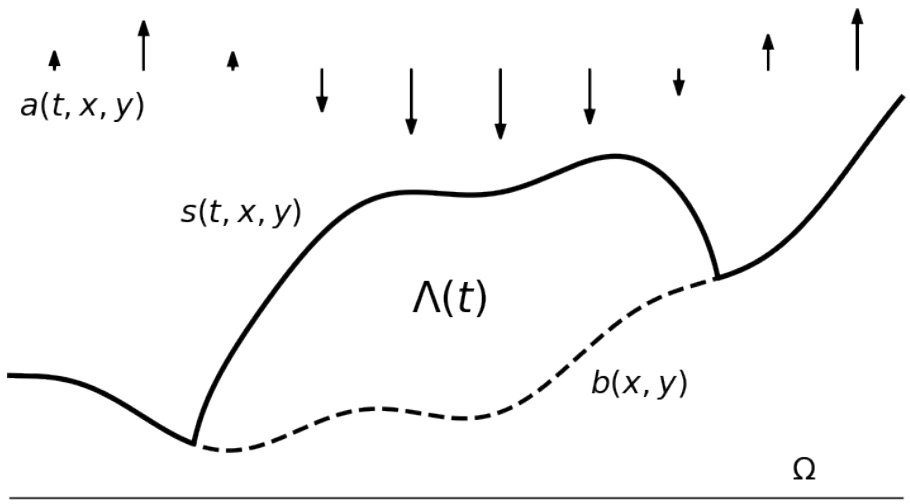
$$-\nabla \cdot (2\nu_\epsilon \, D\mathbf{u}) + \nabla p = \rho \mathbf{g} \qquad \textit{stress balance}$$

$$\nabla \cdot \mathbf{u} = 0 \qquad \textit{incompressibility}$$

- 3 solution variables: s, $\mathbf{u}$, p
- 2 coupling directions:

I. the free-surface equation has $\mathbf{u}|_s$ as a coefficient

II. $s(t, x, y)$ determines the evolving domain $\Lambda(t)$ on which the Stokes problem is solved at each instant



weak form of coupled system of equations

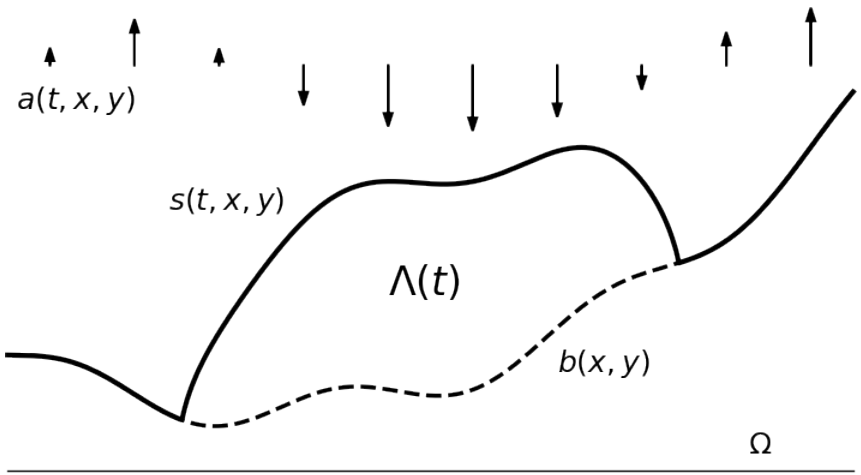
- we get the weak form of the coupled system in the usual way
- multiply by test functions and integrate:

$$\int_\Omega \left(\frac{\partial s}{\partial t} - \mathbf{u}|_s \cdot \mathbf{n}_s - a \right) r \, dx = 0 \qquad \forall r$$

$$\int_{\Lambda(t)} 2\,\nu_\epsilon(|D\mathbf{u}|)\, D\mathbf{u} \,:\, D\mathbf{v} - p\nabla\cdot\mathbf{v} - \rho\mathbf{g}\cdot\mathbf{v}\, dx = 0 \qquad \forall \mathbf{v}$$

$$\int_{\Lambda(t)} -(\nabla\cdot\mathbf{u}) q \, dx = 0 \qquad \forall q$$

- high-level choices for system:
  1. spaces and elements for $s, \mathbf{u}, p$?
  2. time-stepping: explicit or implicit?
  3. solved together (*monolithic*) or split?



challenges in solving the coupled system

- early/naive approaches to solving this coupled problem have been fragile
- the method everyone tries first is to solve power-law Stokes for the current geometry, then advance the surface, and repeat
  - this is an *explicit* approach, which *splits* the coupling
  - resulting performance is stability-limited, in a manner which is not well understood

- what are the underlying challenges in designing better schemes?

1. the mesh used for the $\mathbf{u}, p$ solver, or at least its geometry, changes at every time step
   - this makes monolithic and/or implicit implementations difficult
2. the coupled problem mixes advective and diffusive aspects
   - proportions depending on stress boundary conditions
   - precise time-step restrictions remain unclear
   - expected performance scaling remains unclear (B 2023)
   - but `stage4/` applies recent theoretical progress!

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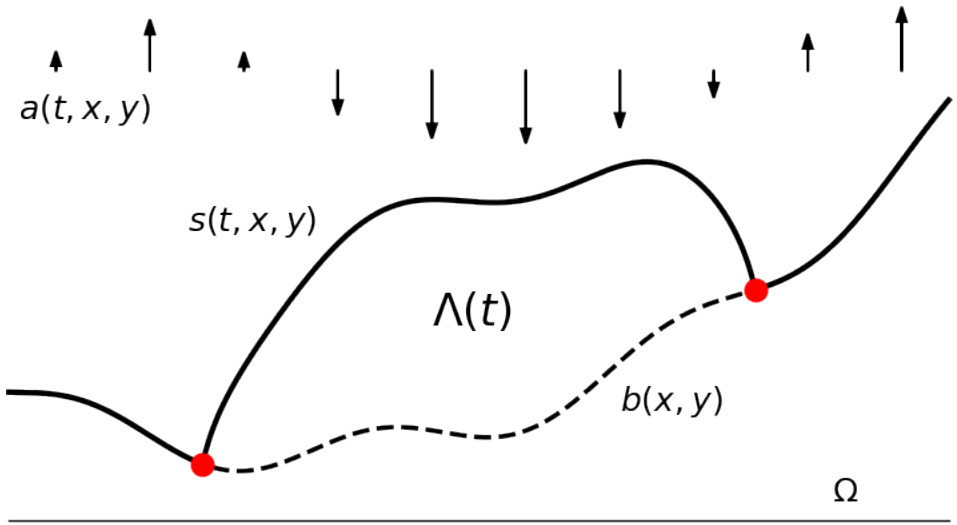
   - but `stage4/` applies recent theoretical progress!

also, glaciers have moving margins!

- the surface elevation must be above the bed
- enforcing this obvious fact provides the (lateral) free-boundary condition for a land-based glacier
  - the weak form is a variational inequality (B 2021b, Chapter 12)
  - the thickness and flux are both zero at the free boundary
- solving the coupled problem subject to this condition determines the location of the moving glacier margin (free boundary)



Outline

theory equations for the glacier dynamics problem

`stage1/` first demo: linear Stokes

`stage2/` power-law Stokes for glacier ice

`stage3/` more power: extruded meshes and 3D glaciers

theory evolving glaciers: free-surface equation coupled to Stokes

`stage4/` moving-margin dome solutions

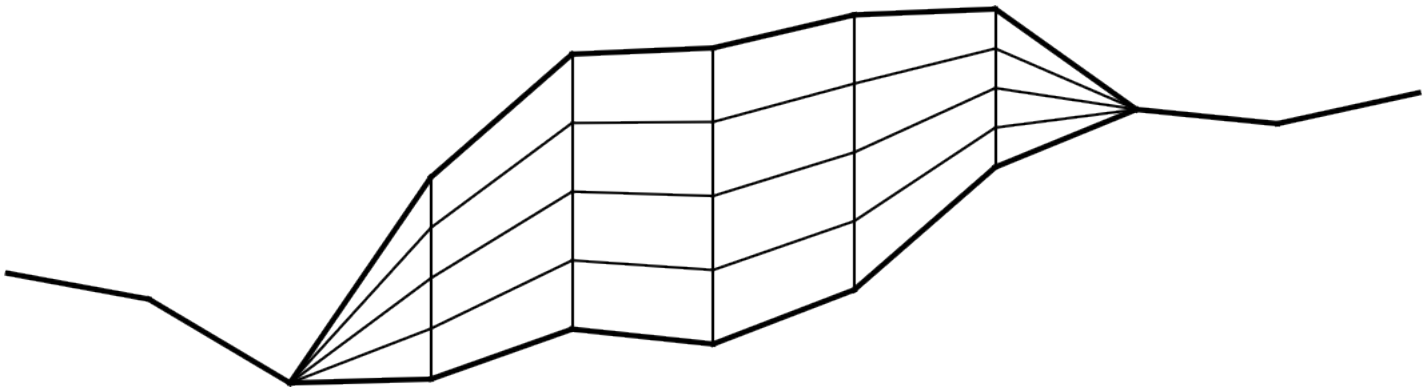
summary and learning more

`github.com/bueler/stokes-ice-tutorial`

evolving extruded meshes

- we will demonstrate and test a 2D solver which is semi-implicit, split, stabilized, and moving-margin
  - . . . as explained on the next few slides
  - caution: “stabilized” will mean several things here

- using an extruded mesh
- between continuous P_1 surfaces $s = s(t, x)$ and $b = b(t, x)$:
  - the Stokes domain $\Lambda(t)$ is polygonal (in space) at each t
  - the elements are quadrilaterals (in 2D)



new techniques for `stage4/`

- our solver is based on these 3 new techniques which modify the weak form of the coupled problem:

1. first-order, and mostly explicit, time-stepping
   A. but we add an implicit edge stabilization in the free-surface equation
   B. and we determine the time step by a CFL condition
2. split, with alternating free-surface and Stokes solves
   - but we add a load stabilization coupling to the Stokes solver
3. the coupled free-surface problem is solved as a variational inequality which determines the glacier extent and margin
   A. we apply a bounds-respecting Newton method
   B. we allow variable ice extent on the extruded mesh
   C. we prefer pressure elements with interior degrees of freedom

- the next slides give citations and details on these techniques
  - techniques 1 and 2 are the “Swedish stabilizations”

technique 1A: implicit edge stabilization in free-surface equation

- this technique is from Tominec et al (2026)
  - see also Ern & Guermond (2013), and refs therein
- add FE edge stabilization, applied semi-implicitly:

$$\int_{\Omega^h} \left(\frac{s^h_{k+1} - s^h_k}{\Delta t} - \mathbf{u}|_{s^h_k} \cdot \mathbf{n}_{s^h_k} - a \right) r^h \, dx + J(s^h_{k+1}, r^h) = 0$$

for all test functions r^h, where

$$J(s, r) = \sum_{e \in E^h} \int_e \tfrac{1}{2} h_e^2 \left| \mathbf{u}|_{s^h_k} \right| [\![\nabla s]\!][\![\nabla r]\!] \, dS$$


Igor Tominec


Lukas Lundgren


Andres Löfgren


Josefin Ahlkrona

- s^h_k is the FE approximation of surface elevation at time t_k
- the system to solve for unknown s^h_{k+1} is linear
- E^h is the set of interior edges in the triangulation Ω^h
- $[\![\mathbf{v}]\!] = \mathbf{v}_+ \cdot \mathbf{n}_+ - \mathbf{v}_- \cdot \mathbf{n}_-$ is a vector field jump along an edge
- h_e is the average of the diameters of the two cells which meet e
- note surface velocity ($\mathbf{u}|_{s^h_k}$) and slope ($\mathbf{n}_{s^h_k}$) are evaluated explicitly, at t_k

technique 1B: CFL determination of time step

- our application adds an advancing glacier margin, which is not considered in the stability proof from Tominec et al (2026)
- edge stabilization technique 1A applies split and semi-implicitly, with velocity evaluated explicitly
  - thus margin advance is limited to one base-mesh element per time step
- the time step Δt is therefore limited by a CFL criterion:

$$\Delta t = C_{\text{CFL}} \min \frac{h}{|\mathbf{u}|}$$

  - we get reasonable results with $C_{\text{CFL}} = 0.25$
    - Ern & Guermond (2013) use $C_{\text{CFL}} = 0.2$ or 0.25 on linear advections with weighted edge stabilization (and SSP3, RK4 schemes)

technique 2: semi-implicit coupling heuristic in stress balance


Andres Löfgren


Josefin Ahlkrona


Christian Helanow

- from Löfgren et al (2022), as refined by Tominec et al (2026)
- add extrapolated surface mass balance load to Stokes form:

$$\int_{\Lambda(t_k)} 2\,\nu_\epsilon(|D\mathbf{u}|)\,D\mathbf{u} \;:\; D\mathbf{v} - p\nabla\cdot\mathbf{v} - \rho\mathbf{g}\cdot\mathbf{v}\,dx$$
$$+ \int_{\Gamma_s(t_k)} \rho g \Delta t \left[\frac{1}{2}(\mathbf{u}|_{s_k}\cdot\mathbf{n}_{s_k}) + a\right](\mathbf{v}\cdot\mathbf{n})\,ds = 0$$

  - $\Lambda(t_k)$ is the Stokes domain defined by the current surface elevation s_k
  - $\Gamma_s(t_k)$ is the upper surface of $\Lambda(t_k)$
  - $\mathbf{n}_{s_k} = \langle -\nabla s_k, 1\rangle$ is an outward normal on $\Gamma_s(t_k)$
  - $\mathbf{n} = \mathbf{n}_{s_k}/|\mathbf{n}_{s_k}|$ is the outward *unit* normal

- concept: modify the stress balance to stabilize the coupled solve
  - Reynold's transport theorem motivates this (symmetrized) extrapolation
  - . . . which puts time step Δt and SMB a inside the Stokes solve!
  - Tominec et al (2026) prove that the L^2 norm of the evolving surface is controlled; it can't blow up

slumping in a box

- *purpose:* exhibit numerical stability and exact mass conservation for an evolving surface between lubricated walls
  - the geometry considered by Löfgren et al (2022), Tominec et al (2026)
- *source files:* `geometry.py`, `halfar.py`, `solve.py` ← inspect these
- *generated file:* `dome.pvd`

```
$ cd stage4/
$ python3 solve.py -walls -L 8000 -omovie.pvd
$ paraview movie.pvd
```

$t = 0$ a:

$\rightarrow$

$t = 5$ a:

instability showing need for stabilization technique 2

- *without* technique 2, the maximum size of the stable time step is significantly restricted (Löfgren et al 2022)
  - many time step choices produce "sloshing" and Stokes solver failures
  - e.g.runs to produce table: `$ python3 solve.py -walls -L 8000 -noload -nocfl -mx MX -maxdt DT` with `MX`$= 25, 50, 100, 200, 400$
- to demonstrate what happens *without* technique 2, for each resolution in this table we started with $\Delta t = 1$ a, and reduced Δt by a factor of two until the 5 a run completes without a solver failure

Δx (m)	stable Δt (a) w/o technique 2
640	0.25
320	0.25
160	0.125
80	0.125
40	0.125

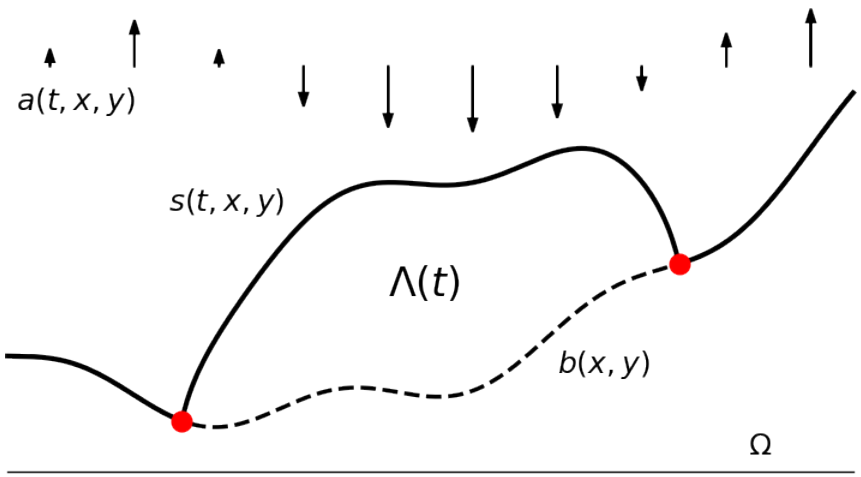
  - Tominec et al (2026) does additional experimentation
- with technique 2, arbitrary time steps are stable, and exhibit no sloshing

technique 3A: semi-implicit variational inequality for free surface

- define the FE admissible set $\mathcal{K}^h = \left\{ r^h \in \mathrm{P}_1 \,:\, r^h \geq b^h \right\}$
  - the P_1 bed elevation function b^h is time-independent
- the new surface $s^h_{k+1} \in \mathcal{K}^h$ solves this variational inequality weak form:

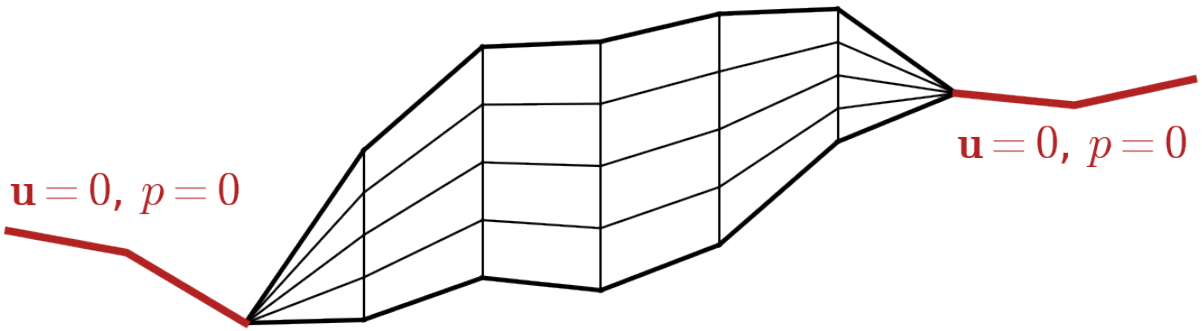
$$\int_\Omega \left(\frac{s^h_{k+1} - s^h_k}{\Delta t} - \mathbf{u}|_{s^h_k} \cdot \mathbf{n}_{s^h_k} - a \right) (r^h - s^h_{k+1})\, dx + J(s^h_{k+1}, r^h - s^h_{k+1}) \geq 0$$

  for all r^h in $\mathcal{K}^h$
  - includes edge stabilization
  - velocity $\mathbf{u}|_{s^h_k}$ still explicit
- the computed margin location is where s^h_{k+1} meets b^h
- we apply the `vinewtonrsls` bounds-respecting Newton solver from PETSc (B 2021b)



technique 3B: extruded mesh with variable ice extent

- the base mesh covers all of Ω, including areas with no ice
- the extruded mesh has a fixed number `mz` of levels in the vertical
  - $\mathbf{u}, p$ are defined at each node of the extruded mesh
  - our meshes and fields have constant memory footprint as we time-step
- in columns with no ice we lock/pinch $\mathbf{u}, p$ values to zero



  - s_k^h, b^h define ice domain $\Lambda(t_k)$ in this semi-implicit framework
  - $s_k^h = b^h$ at a base mesh node $\implies$ no ice in that extruded column
  - `PinchColumnPressure|Velocity` sub-class `DirichletBC`

technique 3C: pressure elements with interior degrees of freedom

- however, we do not want to pinch p to be zero anywhere in the closest-to-margin elements containing ice
  - doing so causes an unintended Dirichlet condition on the pressure
- therefore prefer pressure elements with interior degrees of freedom:

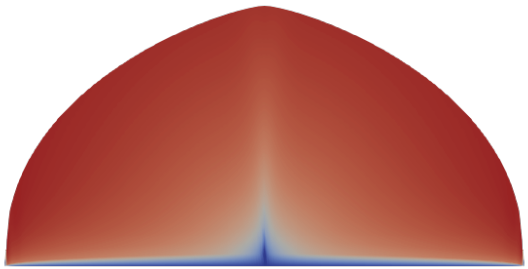
```
W = FunctionSpace(mesh, "DQ", 1, variant="spectral")  # best

W = FunctionSpace(mesh, "DG", 0)  # adequate
```

- for such elements the moving-margin mass conservation error (later slides) is provably one-sided

a dome flowing outward

- *purpose:* moving-margin free surface, exhibiting numerical stability
- *source files:* `geometry.py`, `halfar.py`, `solve.py` ← inspect these
- *generated file:* `movie.pvd`

```
$ python3 solve.py -mx 200 -mz 10 -T 100 -L 20000 \
    -omovie movie.pvd
$ paraview movie.pvd
```



$t = 0$ a

$\rightarrow$

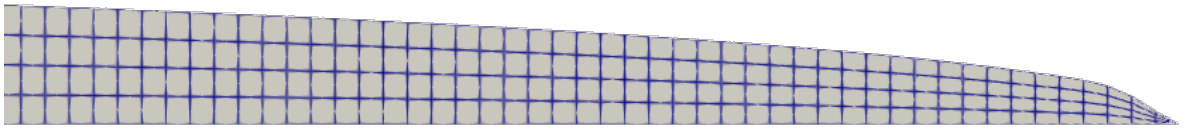
$t = 100$ a

10$\times$ vertical exaggeration, $\log(|\mathbf{u}|)$ coloring

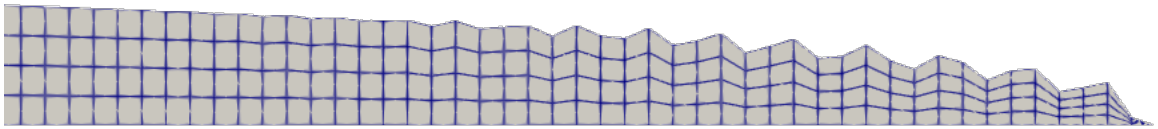
instability showing need for stabilization technique 1A

- especially near the glacier margin, implicit edge stabilization (technique 1A) is clearly needed
- compare these runs:

```
$ python3 solve.py -mx 200 -T 1
```



```
$ python3 solve.py -mx 200 -T 1 -noedge
```

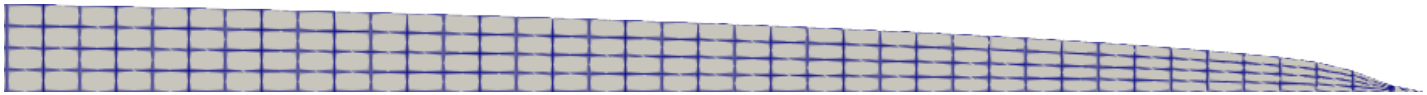


- runs with `-walls` generally do not show the need

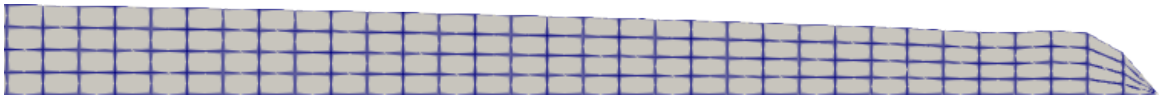
instability showing need for stabilization technique 1B

- need for technique 1B (CFL for time-step length) is more subtle
- compare the margin shape at the end of these two runs:

```
$ python3 solve.py -mx 100
```



```
$ python3 solve.py -mx 100 -nocfl -maxdt 1.0
```

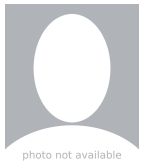


  - advancing margin shape without CFL-limited time steps is wrong
    - bulging and not downsloping
    - not close to Halfar margin location (next slides)
    - but not obviously unstable

assessing correctness

Q. is our time-dependent solver correct?

- it is not clear how to build a moving-margin, coupled Stokes-surface exact solution
- however, such exact solutions exist for shallow ice approximation (SIA)
- SIA exact solutions are close to solving the free-surface + Stokes coupled equations when the aspect ratio $\epsilon = (\text{height})/(\text{width})$ is small
- next slide: we compare with the Halfar (1981) exact solution
  - on a shallow initial dome geometry, SIA is reasonably accurate
  - e.g. initial height 1000 m and width 20 km yields $\epsilon = 0.05$



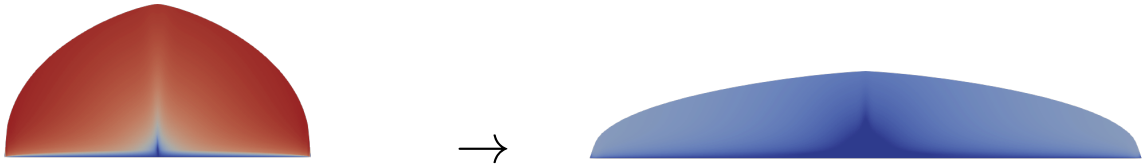
Peter Halfar

evidence of correctness for a shallow, moving-margin dome

- consider runs of 10 years at different horizontal resolutions:

```
$ python3 solve.py -mz 10 -T 10 -mx MX
```

  - use `MX`= 50, 100, 200, 400, 800, giving $75 \le \Delta x \le 1200$ m
  - from 33 (`MX`= 50) to 557 (`MX`= 800) adaptive-scheme time steps
  - max computed ice speed 4600 m/a initially; 160 m/a for final state



- compare final computed geometry to Halfar's prediction:

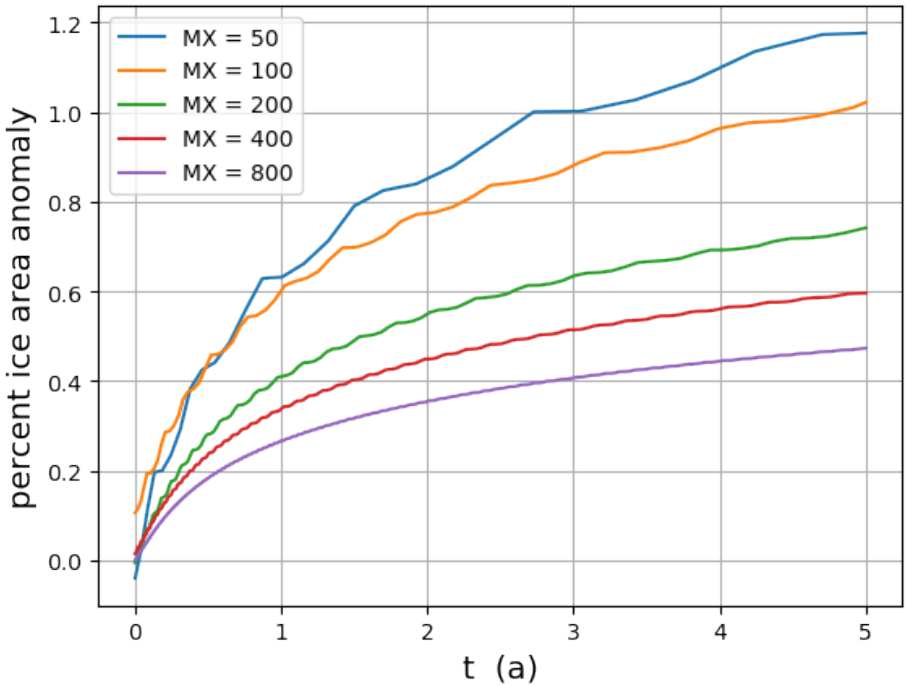
Δx (m)	width (km)	center height (m)
600	28.800	697.789
300	28.800	697.026
150	28.800	696.107
75	28.950	695.670
37.5	28.875	695.329
Halfar	28.993	689.829

- reasonably close, but we do *not* expect convergence to Halfar value

mass conservation errors with moving glacier margins

- since $a = 0$ here, mass is exactly conserved in the continuum model

- unfortunately, our scheme does not conserve mass here
- numerical mass (= ice area) drifts about 1% over 5 years
  - curves should be horizontal lines
  - higher res $\Longrightarrow$ better conserve



- this is only a moving margin effect
  - our scheme conserves mass for fixed walls, as proved by Tominec 2026
  - here mass is non-decreasing; prove from VI using $a = 0$
  - this margin mass conservation issue is distinct from those in Jarosch et al (2013) and B (2021a)

Outline

theory equations for the glacier dynamics problem

`stage1/` first demo: linear Stokes

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`stage3/` more power: extruded meshes and 3D glaciers

theory evolving glaciers: free-surface equation coupled to Stokes

`stage4/` moving-margin dome solutions

summary and learning more

`github.com/bueler/stokes-ice-tutorial`

summary and outlook

- Firedrake makes solving the glacier Stokes problem easy for the new numerical modeler:
  - the Python codes in this tutorial are short
  - UFL (weak forms) and PETSc (nonlinear solvers) are built-in to the stack

- modern techniques work:
  - extruded meshes (Firedrake), because glaciers are thin flows
  - Swedish stabilizations, so semi-implicit time-stepping is safe and easy
    - Tominec et al (2026) has 1st coupled Stokes-surface *proof* (to my knowledge)
  - bounds-constrained Newton solvers (PETSc), giving moving margins

- remaining challenges and improvements:
  1. make the Stokes solver faster by replacing the direct linear algebra with a scalable matrix-free solver (presumably multigrid)
  2. make the simulation faster by combining the free-surface equation and the Stokes solver in a monolithic implicit step solved by a scalable solve (presumably multigrid $=$ FASCD?)
  3. make time-stepping glacier simulations mass conserving even in free-boundary cases

learn more

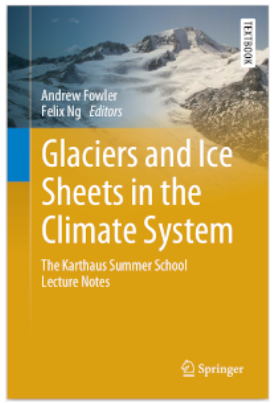
`github.com/bueler/stokes-ice-tutorial`

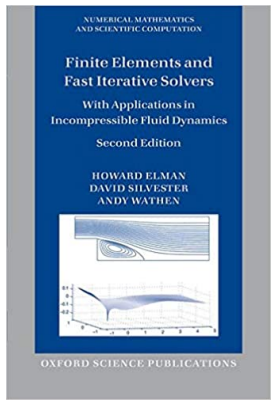
- Firedrake: `firedrakeproject.org`
- PETSc: `petsc.org`

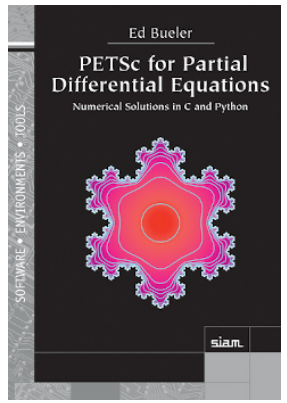
Firedrake

PETSc

- for power-law Stokes eqns, see Ch. 1 by Hewitt in Fowler & Ng (2021)
- for finite elements and Stokes: Elman, Silvester, & Wathen (2014)
- for PETSc, Firedrake, and Stokes: Bueler (2021b)







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